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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » The Joy of Computing using Python (course)

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Week 6 : Assignment 6

The due date for submitting this assignment has passed.

Due on 2025-03-05, 23:59 IST.

Assignment submitted on 2025-03-05, 22:02 IST

1) If n is a positive integer, what is the output of the function given input n, ***1 point***

```
def mystery(n):
  if n <= 0:
    return 0
  return n + mystery(n - 1)
```

- (selected) Sum of numbers from 1 to n
- Sum of numbers from 1 to n-1
- n-1
- n

Yes, the answer is correct.
Score: 1

Accepted Answers:
Sum of numbers from 1 to n

2) **Assertion (A):** If a recursive function doesn't have a base case, it may never stop. ***1 point***
Reason (R): Recursion means that a function calls itself.

- Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (selected) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (A) is true, but (R) is false.
- (A) is false, but (R) is true.

Yes, the answer is correct.
Score: 1

Accepted Answers:
Both (A) and (R) are true, but (R) is not the correct explanation of (A).

3) What is the output of following code ? ***1 point***

```
def func(x):
  return x ** 2

def func(x, y=1):
  return x**2 + y

print(func(5))
```

- Error
- 25

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

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Week 2 ()

Introduction to Anaconda (unit?unit=37&lesson=38)

Installation of Anaconda (unit?unit=37&lesson=39)

Introduction to Spyder IDE (unit?unit=37&lesson=40)

Printing statements in Python (unit?unit=37&lesson=41)

Understanding Variables in Python (unit?unit=37&lesson=42)

Executing a sequence of instructions in the Console (unit?unit=37&lesson=43)

Writing your First Program (unit?unit=37&lesson=44)

Taking inputs from the user (unit?unit=37&lesson=45)

Discount Calculation (unit?unit=37&lesson=46)

Motivation to if condition (unit?unit=37&lesson=47)

A reminder on how to deal with numbers (unit?unit=37&lesson=48)

Understanding if condition's working (unit?unit=37&lesson=49)

- Realizing the importance of syntax and indentation (unit?unit=37&lesson=50)
- Introductions to loops (unit?unit=37&lesson=51)
- Loops: Sum of numbers (unit?unit=37&lesson=52)
- Loops: Sum of numbers (continued) (unit?unit=37&lesson=53)
- Loops: Multiplication Tables (unit?unit=37&lesson=54)
- Introduction to While Loop (unit?unit=37&lesson=55)
- Week 2 Feedback Form: The Joy of Computing using Python (unit?unit=37&lesson=56)
- Quiz: Week 2 : Assignment 2 (assessment?name=503)
- Week 2: Programming Assignment 1 (/noc25_cs69/progassignment?name=504)
- Week 2: Programming Assignment 2 (/noc25_cs69/progassignment?name=505)
- Week 2: Programming Assignment 3 (/noc25_cs69/progassignment?name=506)

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

- Substitution Cipher -The science of secrecy (unit?unit=124&lesson=125)
- Substitution Cipher -The science of secrecy 01 (unit?unit=124&lesson=126)
- Substitution Cipher -The science of secrecy 02 (unit?unit=124&lesson=127)
- Substitution Cipher -The science of secrecy 03 (unit?unit=124&lesson=128)
- Tic Tac Toe - Down the memory Lane (unit?unit=124&lesson=129)
- Tic Tac Toe - Down the memory Lane 01 (unit?unit=124&lesson=130)
- Tic Tac Toe - Down the memory Lane 02 (unit?unit=124&lesson=131)

- (selected) 26
- 5

Yes, the answer is correct.
Score: 1

Accepted Answers:
26

4) In English, the letter 'z' usually appears far less often than most letters. Now imagine applying a Substitution Cipher to a large body of English text, where 'z' is replaced by 'r', and every other letter is also mapped to a unique letter. After encryption, which letter is likely to be the least frequent in the new text? ***1 point***
Hint: Search the internet for more info, if needed.

- z
- (selected) r
- e
- It is not possible to determine

Yes, the answer is correct.
Score: 1

Accepted Answers:
r

5) If you have a large ciphertext produced by a simple substitution cipher on an English text, can you determine the original message by examining how often each letter appears? ***1 point***
Hint: Search the internet for more info, if needed.

- (selected) Yes, it can be done.
- No, it cannot be done.

Yes, the answer is correct.
Score: 1

Accepted Answers:
Yes, it can be done.

6) Suppose you suspect that a piece of ciphertext was produced using a simple sub-stitution cipher. Under which circumstances might frequency analysis fail to yield accurate results? ***1 point***

- (selected) The ciphertext is short.
- (selected) The ciphertext was encrypted with another method before encrypting with sub-stitution cipher.
- The ciphertext is very long.

Yes, the answer is correct.
Score: 1

Accepted Answers:
The ciphertext is short.
The ciphertext was encrypted with another method before encrypting with sub-stitution cipher.

7) If variable **dict_name** is a non-empty dictionary, what does dict name.values() re-turn? ***1 point***

- Returns nothing, but prints all the keys in the dictionary.
- Returns a list of all the keys in the dictionary.
- (selected) Returns a list of all the values in the dictionary.
- Returns a list of all the items in the dictionary.

Yes, the answer is correct.
Score: 1

Accepted Answers:
Returns a list of all the values in the dictionary.

8) **Assertion (A):** A Caesar Cipher is a type of Substitution Cipher. ***1 point***
Reason (R): In a Caesar Cipher, each letter in the plaintext is replaced by a letter at a fixed shift in the alphabet.

- (selected) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (A) is true, but (R) is false.
- (A) is false, but (R) is true.

Yes, the answer is correct.
Score: 1

Accepted Answers:
Both (A) and (R) are true, and (R) is the correct explanation of (A).

9) Which outcome generally follows from providing a proper base case in a recursive function? ***1 point***

- (selected) The function returns a predefined result when the base condition is met.
- (selected) It prevents infinite recursion and stops when a specific codition is met.

Tic Tac Toe - Down the memory Lane 03 (unit?unit=124&lesson=132)
Tic Tac Toe - Down the memory Lane 04 (unit?unit=124&lesson=133)
Tic Tac Toe - Down the memory Lane 05 (unit?unit=124&lesson=134)
Recursion (unit?unit=124&lesson=135)
Recursion 01 (unit?unit=124&lesson=136)
Recursion 02 (unit?unit=124&lesson=137)
Recursion 03 (unit?unit=124&lesson=138)
Recursion 04 (unit?unit=124&lesson=139)
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It ensures recursion depth is always constant.

It allows the function to keep calling itself indefinitely.

Yes, the answer is correct.
Score: 1

Accepted Answers:
The function returns a predefined result when the base condition is met.
It prevents infinite recursion and stops when a specific codition is met.

10) In a standard 3×3 Tic-Tac-Toe board, how many distinct winning lines (rows, columns, or diagonals) pass through the center square? ***1 point***

3

2

5

4

Yes, the answer is correct.
Score: 1

Accepted Answers:
4